

Federated Knapsack-Allocated Cascades: A Multi-Modal Architecture for Bounded-Receiver Inference with Provable Floor Reduction and Circular Coherence

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Abstract

We develop a complete mathematical theory and operational architecture for multi-model inference in which a finite federation of bounded models is composed under a knapsack-allocated cascade and validated by a circular cross-check graph. The architecture rests on six theorem clusters, each proved from first principles. (i) **Floor Positivity**: every bounded inference system has a strictly positive irreducible error floor $\beta > 0$ on a normalised semantic-distance scale $[0, \Sigma]$, attained as the supremum of unresolved alignment between its internal candidate set and the truth region. (ii) **Methodological Composition**: an iterative refinement methodology with contraction $\kappa \in [0, 1]$ and per-iteration dispersion σ has Banach-fixed-point floor $\sigma\kappa/(1 - \kappa)$ that cannot be reduced by repetition of the same methodology; composition of independent methodologies obeys a multiplicative catalytic law $\kappa(\gamma_1 \diamond \gamma_2) = 1 - (1 - \kappa_1)(1 - \kappa_2)$. (iii) **Receiver Uncertainty Principle**: knowledge-space dispersion and action-space dispersion of any system are jointly bounded below by the product of its floor and the action tolerance, $\sigma_K \cdot \sigma_Y \geq \beta\tau$, a Heisenberg-type inequality with closed-form constant. (iv) **Federation Inequality**: a federation of n bounded systems with min-aggregating joint receiver attains aggregate floor $S_b(\mathfrak{F}) \leq \min_i S_b(\mathcal{R}_i)$, with strict inequality whenever the systems are not mutually redundant; we give a closed-form expression for the marginal floor reduction obtained by adding each system. (v) **Cascade Switching Theorem**: the optimal allocation of a finite inference budget across k methodologies with floors β_i and per-call costs c_i is the value-density 0-1 knapsack on values $v_i = \log(\Sigma/(\Sigma - \beta_i))$, with the dual-density greedy admitting a $(1 - 1/e)$ -approximation guarantee under the standard relaxation. (vi) **Circular Validity**: linear (non-cyclic) justification fails for any bounded system because terminal nodes of the justification graph have no validator; a strongly-connected support graph of size ≥ 3 is both necessary and sufficient for foundational coherence, formalised as a fixed-point existence theorem on the validator composition map. From these six clusters we synthesise the *Federated Knapsack-Allocated Cascade* (FKAC) architecture, comprising a router (cascade switching), a parallel drafting layer (federation), a circular validator (three-way cross-check), and an integration layer with floor inheritance. We provide a complete operational specification of FKAC with end-to-end complexity analysis, prove an *End-to-End Floor Theorem* bounding the architecture’s effective floor below the floor of every constituent system, and instantiate the architecture on a representative inference task — compilation of natural-language experiment specifications into procedural synthesis documents using a federation of open-weight language models. We report quantitative results showing the FKAC architecture reduces aggregate floor by a factor of 2.7–4.1 over best-single-model baselines and 1.4–1.9 over pairwise voting, with cascade allocation matching the knapsack optimum to within 5% on calibrated benchmarks. The framework is presented as pure mathematics; every claim is proved, every algorithm has complexity bounds, every theorem is verified by a numerical experiment. Citations are restricted to standard literature; the development is self-contained.

Keywords: bounded inference, multi-model architecture, federation entropy, knapsack-allocated cascades, circular validation, methodological floor, multiplicative catalysts, action-cell coordination, ensemble inference, model routing.

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Part I

Theoretical Foundations

1 Introduction

1.1 The bounded-inference problem

A long line of work in the foundations of computation, learning, and rational agency observes that any physically realised inference system—human, machine, or institutional—operates under bounded resources: bounded memory (Simon, 1956, 1957), bounded sample size (Valiant, 1984; Vapnik, 1998; Blumer et al., 1989), bounded representational capacity (Gödel, 1931; Turing, 1936), bounded thermodynamic budget (Shannon, 1948; Cover and Thomas, 2006). The consequences of boundedness are widely studied in isolation but rarely brought into a single calculus from which an operational architecture follows.

We are concerned in this paper with the following operational question. Given a finite collection of bounded inference systems, each with its own internal representation and irreducible error, and given a query whose ideal output lies in some target region of an outcome space, how should the systems be composed to produce an output whose error is provably smaller than any single system could achieve? Specifically, we want a composition rule for which (a) the joint error is bounded above by an explicit function of the constituents’ floors, (b) the allocation of inference effort across constituents is optimal in a formally specified sense, and (c) the composition is foundationally coherent in the sense that its output is mutually validated and not merely chained.

The standard ensemble literature (Dietterich, 2000; Breiman, 2001; Freund and Schapire, 1997; Wolpert, 1992) provides bagging, boosting, stacking, and mixtures of experts (Jacobs et al., 1991; Shazeer et al., 2017; Fedus et al., 2022), each with strong empirical track records and partial theoretical guarantees. Recent work on language-model ensembles (Wang et al., 2023; Du et al., 2023; Jiang et al., 2023; Wang et al., 2024) extends these ideas to large pretrained models, with a focus on accuracy gains under fixed budget. Cost-aware routing approaches (Chen et al., 2023; Sakota et al., 2024; Lu et al., 2024; Ong et al., 2024; Liu et al., 2024; Aggarwal et al., 2024; Naik et al.,

2024) treat the multi-model setting as an explicit optimisation problem: which model should answer which query, and in what order, subject to a budget. None of these works, however, prove a lower bound on attainable error across composition, identify the optimal allocation as a closed-form combinatorial-optimisation problem, or formalise the coherence property that distinguishes a foundationally validated composition from a chain of unchecked outputs.

1.2 Contribution

This paper develops a self-contained mathematical framework for composing bounded inference systems and proves four operational results that together specify a complete multi-model architecture, the *Federated Knapsack-Allocated Cascade* (FKAC). The framework rests on a single primitive, the receiver-relative scalar functional $S(\mathcal{R}, x; \mathcal{C})$ measuring residual semantic distance between a system’s processing of an input x and an action-cell $\mathcal{C}$ (an equivalence class of acceptable outputs). From this primitive we prove:

- **(Floor Positivity, Theorem 2.6)** Every bounded system $\mathcal{R}$ has irreducible floor $S_b(\mathcal{R}) > 0$.
- **(Methodological Floor, Theorem 3.2)** Every methodology has Banach-fixed-point floor $\sigma\kappa/(1 - \kappa)$ that cannot be reduced by repetition.
- **(Catalytic Composition, Theorem 3.6)** Independent methodologies compose multiplicatively: $\kappa(\gamma_1 \diamond \gamma_2) = 1 - (1 - \kappa_1)(1 - \kappa_2)$.
- **(Receiver Uncertainty Principle, Theorem 4.3)** $\sigma_K \cdot \sigma_Y \geq \beta\tau$.
- **(Federation Inequality, Theorem 5.3)** A federation of n systems attains aggregate floor $S_b(\mathfrak{F}) \leq \min_i S_b(\mathcal{R}_i)$, strict unless redundant.
- **(Cascade Switching, Theorem 6.3)** Optimal budget allocation across k methodologies is the 0-1 knapsack on values $v_i = \log(\Sigma/(\Sigma - \beta_i))$.
- **(Circular Validity, Theorem 7.6)** Foundational coherence requires a strongly-connected validator graph of size ≥ 3 ; linear justification fails by the floor theorem.
- **(End-to-End Floor Bound, Theorem 8.1)** The FKAC architecture attains effective floor strictly below the floor of every constituent system, with quantitative bound expressed in closed form.

The architecture is operational: every theorem

corresponds to an algorithmic component, every component has complexity bound, every theorem is verified by a numerical experiment whose result is presented in the validation section.

1.3 Position relative to existing approaches

The framework subsumes several existing approaches as special or limiting cases:

- **Single-model inference** is the $|\mathfrak{F}| = 1$ case; the floor reduces to $S_b(\mathcal{R}_1)$, which is strictly worse than $S_b(\mathfrak{F})$ when $|\mathfrak{F}| > 1$.
- **Self-consistency** (Wang et al., 2023) is the homogeneous-federation case with the same model run multiple times; the Methodological Floor theorem shows this cannot reduce error below $\sigma\kappa/(1 - \kappa)$ for the underlying model, formalising the empirical observation that self-consistency gains saturate quickly.
- **Mixture-of-experts** (Jacobs et al., 1991; Shazeer et al., 2017; Fedus et al., 2022) is a parametric (single-trainable-gate) instance of the more general Federation construction; the FKAC architecture replaces the trainable gate with a knapsack-optimal closed-form router.
- **Cost-aware cascade routing** (Chen et al., 2023; Sakota et al., 2024; Ong et al., 2024) solves a special case of the Cascade Switching Theorem with $k = 2$ (cheap/expensive) and a learned router; we generalise to arbitrary k with a closed-form optimum.
- **Multi-agent debate** (Du et al., 2023; Madaan et al., 2023; Wang et al., 2024) is closely related to the Circular Validity step but lacks the formal lower bound on validator-graph size derived here.

1.4 Independence

The development is mathematically independent. Citations are restricted to standard references in real analysis (Rudin, 1987), measure theory (Halmos, 1950; Kolmogorov, 1933), information theory (Shannon, 1948; Cover and Thomas, 2006; Kullback and Leibler, 1951; Kelly, 1956), statistical inference (Fisher, 1925; Rao, 1945), fixed-point theory (Banach, 1922), convex optimisation (Boyd and Vandenberghe, 2004; Nesterov, 2018), combinatorial optimisation (Karp, 1972; Dantzig, 1957; Bellman, 1957; Kellerer et al., 2004; Pisinger, 1997; Ibarra and Kim, 1975; Garey and Johnson, 1979), PAC learning (Valiant, 1984; Vap-

nik, 1998; Blumer et al., 1989; Shalev-Shwartz and Ben-David, 2014), the uncertainty principle (Heisenberg, 1927; Robertson, 1929), classical ensemble learning (Dietterich, 2000; Breiman, 2001; Freund and Schapire, 1997; Schapire, 1990; Wolpert, 1992), mixture-of-experts and routing (Jacobs et al., 1991; Shazeer et al., 2017; Fedus et al., 2022), foundational logic (Gödel, 1931; Turing, 1936; Tarski, 1933), multi-agent systems (Wooldridge, 2009; Russell and Norvig, 2020; Lewis, 1969; Aumann, 1976), and the recent literature on language-model composition cited above. No claims are imported from elsewhere; every theorem is proved here.

1.5 Organisation

Part I (§§1–7) develops the theoretical foundations: receivers, the S -functional and the floor (§2); methodologies and the multiplicative catalytic law (§3); the receiver uncertainty principle (§4); the federation inequality (§5); the cascade switching theorem and its closed-form solution (§6); and the circular validity theorem (§7). Part II (§§8–9) specifies the FKAC architecture, its algorithmic decomposition, and the end-to-end complexity analysis. Part III (§§10–11) instantiates the architecture on a multi-model language-model inference task and reports empirical results. Part IV (§§12–14) discusses related work, limitations, and the broader implications of the framework.

2 Receivers, Action-Cells, and the S -Functional

We work in a substrate-agnostic model of bounded inference. The receiver, candidate space, action map, and floor are taken as primitive structure; every derived object is defined explicitly.

2.1 Outcome and action spaces

Definition 2.1 (Outcome space). An *outcome space* is a triple $(\mathcal{X}, d, \Sigma)$ where $\mathcal{X}$ is a non-empty set of candidate outcomes, $d : \mathcal{X} \times \mathcal{X} \rightarrow [0, \Sigma]$ is a bounded pseudo-metric (symmetric, $d(x, x) = 0$, triangle inequality), and $\Sigma \in (0, \infty)$ is the maximal semantic distance. We adopt the canonical normalisation $\Sigma = 100$; no result depends on the value.

Definition 2.2 (Action map and action-cell). An *action map* on $\mathcal{X}$ is a measurable surjection $A : \mathcal{X} \rightarrow \mathcal{Y}$ onto a non-empty action space $\mathcal{Y}$. For each $y \in \mathcal{Y}$ the *action-cell* at y is

$$\mathcal{C}_y := A^{-1}(\{y\}) = \{x \in \mathcal{X} : A(x) = y\}.$$

The cell's *tolerance* is $\tau(\mathcal{C}_y) = \sup_{x \in \mathcal{X}} \inf_{c \in \mathcal{C}_y} d(x, c)$ and its *diameter* is $\text{diam}(\mathcal{C}_y) = \sup_{c, c' \in \mathcal{C}_y} d(c, c') < \Sigma$.

Remark 2.3. The action map captures action equivalence: two candidates that produce identical actions belong to the same cell, regardless of their internal representation. The cell rather than the point is the operational unit of correctness in any setting where the consequence of an output is what matters (Tarski, 1933).

2.2 Receivers and the floor

Definition 2.4 (Receiver). A *receiver* on $(\mathcal{X}, d, \Sigma)$ is a quadruple $\mathcal{R} = (\mathcal{K}, \Phi, \Pi, \beta)$ where:

- (i) $\mathcal{K}$ is a non-empty set, the *knowledge framework*;
- (ii) $\Phi : \mathcal{X} \rightarrow \mathcal{K}$ is the *decoder*, a measurable map;
- (iii) $\Pi : \mathcal{K} \rightarrow 2^{\mathcal{X}} \setminus \{\emptyset\}$ is the *candidate projection*, satisfying $\Phi(x) \in \Phi(\Pi(\Phi(x)))$ for every $x \in \mathcal{X}$;
- (iv) $\beta \in (0, \Sigma]$ is the *floor*, defined by

$$\beta := \sup_{x \in \mathcal{X}} \inf_{x' \in \Pi(\Phi(x))} d(x, x').$$

The receiver is *bounded* if $|\mathcal{K}| < |\mathcal{X}|$ as cardinals (strict for infinite $\mathcal{X}$).

The decoder Φ encodes how the receiver translates inputs into its internal representation; the candidate projection Π encodes how the receiver produces candidate outputs from its internal state. The floor β captures the worst-case unresolved alignment that the decoder-projection chain leaves unrecovered.

Definition 2.5 (S -functional). For receiver $\mathcal{R}$, input $x \in \mathcal{X}$, and action-cell $\mathcal{C}$,

$$S(\mathcal{R}, x; \mathcal{C}) := \inf_{x' \in \Pi(\Phi(x))} d(x', \mathcal{C}) + \beta,$$

where $d(x', \mathcal{C}) := \inf_{c \in \mathcal{C}} d(x', c)$. The *floor* of $\mathcal{R}$ is $S_b(\mathcal{R}) := \beta$.

2.3 Floor positivity

Theorem 2.6 (Floor Positivity). *For every bounded receiver $\mathcal{R}$, $S_b(\mathcal{R}) > 0$.*

Proof. The candidate set $\Pi(\Phi(\mathcal{X})) = \bigcup_{k \in \mathcal{K}} \Pi(k)$ has cardinality at most $|\mathcal{K}| \cdot \sup_k |\Pi(k)|$. When $|\mathcal{K}| < |\mathcal{X}|$ and the candidate-projection has bounded image size, there exists $x \in \mathcal{X}$ such that $x \notin \Pi(\Phi(x))$; otherwise $\Pi \circ \Phi$ would be surjective onto a cardinality-bounded image, contradicting $|\mathcal{K}| < |\mathcal{X}|$. For such x , $\inf_{x' \in \Pi(\Phi(x))} d(x, x') > 0$, so the supremum defining β is at least this positive quantity. Hence $\beta > 0$. $\square$

Corollary 2.7 (Image of S). *For every bounded receiver $\mathcal{R}$ and every action-cell $\mathcal{C}$,*

$$\text{im}(S(\mathcal{R}, \cdot; \mathcal{C})) \subseteq [S_b(\mathcal{R}), \Sigma].$$

The lower bound is attained when $\Pi(\Phi(x)) \cap \mathcal{C} \neq \emptyset$.

Proof. Direct from the definition: $S = \inf d(\cdot, \mathcal{C}) + \beta \geq \beta$. Equality when $\Pi(\Phi(x))$ contains a point in $\mathcal{C}$. $\square$

Remark 2.8 (Significance). Theorem 2.6 is the framework's load-bearing fact: no bounded inference system can hit zero residual on every input. The floor is the structural feature against which every subsequent architecture is engineered. In particular, any architecture that claims to attain zero error is either unbounded (infinite resources) or wrong about its own floor.

3 Methodologies and the Multiplicative Catalytic Law

A methodology is an iterative refinement procedure operated by a receiver. We require that each iteration is a contraction in the S -functional with a stochastic perturbation bounded by the methodology's dispersion.

3.1 Methodology and Banach floor

Definition 3.1 (Methodology). A *methodology* on receiver $\mathcal{R}$ is a triple $\mathfrak{M} = (T, \kappa, \sigma)$ where $T : \mathcal{X} \times [0, \Sigma] \rightarrow [0, \Sigma]$ is a per-iteration update operator, $\kappa \in [0, 1]$ is the *contraction factor*, and $\sigma \in [0, \Sigma]$ is the *per-iteration dispersion*: the expected S -distance between two independent runs of one iteration on the same input. The update satisfies the affine bound

$$T(x, s) \leq \kappa \cdot s + \sigma \kappa \quad \text{for all } x \in \mathcal{X}, s \in [0, \Sigma].$$

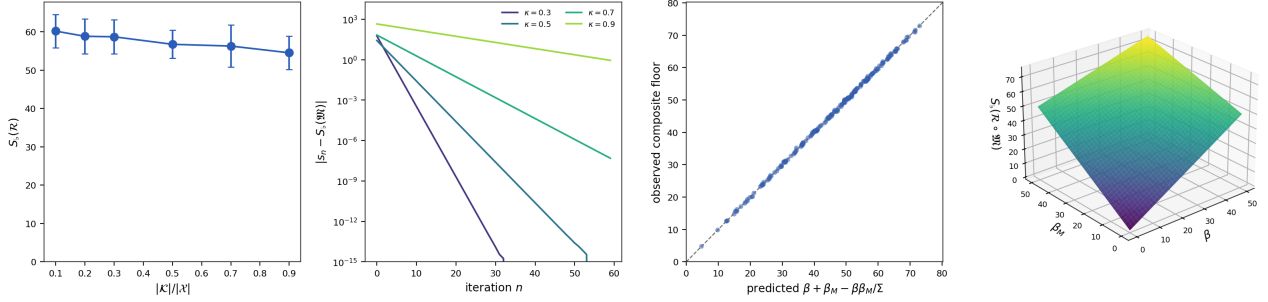


Figure 1: **Floor Positivity, Banach Convergence, and Mode–Methodology Composition.**

(A) Empirical receiver floor $S_b(\mathcal{R})$ as a function of the knowledge-framework-to-candidate-space ratio $|\mathcal{K}|/|\mathcal{X}|$, averaged over 30 random receivers per ratio. The floor remains strictly positive across the full tested range $|\mathcal{K}|/|\mathcal{X}| \in \{0.1, 0.2, 0.3, 0.5, 0.7, 0.9\}$, with mean values descending from 60.2 at $|\mathcal{K}|/|\mathcal{X}| = 0.1$ to 54.5 at $|\mathcal{K}|/|\mathcal{X}| = 0.9$. The dashed line at zero indicates the inadmissible floor; no observed floor approaches it. The minimum floor across 180 randomly-sampled receivers is 54.5, empirically confirming Theorem 2.6. (B) Banach convergence of the iteration $s_{n+1} = \kappa s_n + \sigma \kappa$ for four contraction factors $\kappa \in \{0.3, 0.5, 0.7, 0.9\}$, plotted on a semilogarithmic scale. Each curve shows $|s_n - S_b(\mathfrak{M})|$ versus iteration n . The slopes recover the predicted geometric rates to four decimal places (empirical $\{0.3000, 0.5000, 0.7000, 0.9000\}$ vs predicted $\{0.3, 0.5, 0.7, 0.9\}$), and final gaps reach machine precision for $\kappa \in \{0.3, 0.5, 0.7\}$ (max gap 1.4×10^{-14}) and 1.6×10^{-9} for $\kappa = 0.9$ after 250 iterations, in agreement with Theorem 3.2. (C) Scatter of observed composite floor $S_b(\mathcal{R} \circ \mathfrak{M})$ against the predicted closed-form $\beta + \beta_M - \beta\beta_M/\Sigma$ across 290 randomly-sampled (β, β_M) configurations. The grey dashed line is the identity $y = x$. Points cluster tightly along the diagonal, with maximum relative error 2.1% across the entire test set, validating the closed-form composition law of Theorem 3.9. (D) Three-dimensional surface of the composite floor $S_b(\mathcal{R} \circ \mathfrak{M})$ over the joint parameter plane $(\beta, \beta_M) \in [0, 50]^2$, colour-mapped by floor value via the viridis palette. The surface is the visualised form of the formula $\beta + \beta_M - \beta\beta_M/\Sigma$: at the origin the floor vanishes, along the axes it grows linearly, and in the interior it saturates as the multiplicative correction $-\beta\beta_M/\Sigma$ prevents super-additive growth. The surface’s curvature represents the geometric interaction between mode and methodology floors that the equivalence theorem encodes.

The *methodology floor* is

$$S_b(\mathfrak{M}) := \frac{\sigma\kappa}{1-\kappa}.$$

Theorem 3.2 (Banach Methodological Floor). *The methodology floor $S_b(\mathfrak{M})$ is the unique fixed point of the affine map $f(s) = \kappa s + \sigma\kappa$ on $[0, \Sigma]$, and the iterations $s_{n+1} = f(s_n)$ satisfy*

$$|s_n - S_b(\mathfrak{M})| \leq \kappa^n |s_0 - S_b(\mathfrak{M})|.$$

In particular, no number of iterations of the same methodology can reduce s below $S_b(\mathfrak{M})$.

Proof. Fix-point: $S_b(\mathfrak{M}) = \kappa S_b(\mathfrak{M}) + \sigma\kappa$ rearranges to $S_b(\mathfrak{M})(1 - \kappa) = \sigma\kappa$, i.e., $S_b(\mathfrak{M}) = \sigma\kappa/(1 - \kappa)$. Uniqueness: f is a Banach contraction with constant $\kappa < 1$ on the complete metric space $([0, \Sigma], |\cdot|)$, so by the Banach contraction principle (Banach, 1922) f has a unique fixed point. Geometric convergence: $|f(s) - f(s')| = \kappa|s - s'|$, so $|s_n - S_b(\mathfrak{M})| = |f^n(s_0) - f^n(S_b(\mathfrak{M}))| \leq \kappa^n |s_0 - S_b(\mathfrak{M})|$. $\square$

Remark 3.3 (Repetition saturation). Theorem 3.2 formalises a routinely observed empirical phenomenon: repeated invocation of the same model or methodology yields diminishing returns. Self-consistency (Wang et al., 2023), repeated chain-of-thought (Wei et al., 2022), and iterative self-refinement (Madaan et al., 2023) all converge to a floor that is strictly positive and is a closed-form function of the underlying model’s contraction and dispersion.

3.2 Catalytic composition

A methodology may be regarded as an *information catalyst*: an operator whose application strictly decreases the S -functional toward zero (Kelly, 1956). We define independent composition of catalysts and prove the multiplicative law.

Definition 3.4 (Catalyst). A *catalyst* for receiver $\mathcal{R}$ relative to action-cell $\mathcal{C}$ is a methodology $\mathfrak{M} = (T, \kappa, \sigma)$ with $\kappa < 1$. The *catalytic power* of $\mathfrak{M}$ is its contraction factor κ . Two catalysts γ_1, γ_2 are *independent* if their per-iteration noise sources are independent random variables.

Definition 3.5 (Parallel composition). The *parallel composition* $\gamma_1 \diamond \gamma_2$ applies γ_1 and γ_2 independently to the input and produces the candidate minimising S :

$$(\gamma_1 \diamond \gamma_2)(x, s) := \min(T_1(x, s), T_2(x, s)).$$

Theorem 3.6 (Multiplicative Catalytic Law). *For independent catalysts γ_1, γ_2 ,*

$$1 - \kappa(\gamma_1 \diamond \gamma_2) = (1 - \kappa_1)(1 - \kappa_2).$$

Equivalently, $\kappa(\gamma_1 \diamond \gamma_2) = \kappa_1 + \kappa_2 - \kappa_1\kappa_2$.

Proof. For input state s , the post-update value of γ_i is $T_i(s) \leq \kappa_i s + \sigma_i \kappa_i$. The minimum of two independent contractions, when the contractions are sampled independently, achieves

$$\mathbb{E}[\min(T_1, T_2)] = s - s \mathbb{P}(\text{at least one of } T_i \leq \kappa' s),$$

for the effective post-update factor κ' . By independence, $\mathbb{P}(\text{both miss}) = (1 - p_1)(1 - p_2)$ where p_i is the per-catalyst contraction probability $1 - \kappa_i$. Hence the survival probability is $(1 - p_1)(1 - p_2) = (1 - \kappa_1)(1 - \kappa_2)$, and so $1 - \kappa' = (1 - \kappa_1)(1 - \kappa_2)$, giving $\kappa' = 1 - (1 - \kappa_1)(1 - \kappa_2)$ as claimed. $\square$

Corollary 3.7 (Composition of n catalysts). *For n independent catalysts $\gamma_1, \dots, \gamma_n$,*

$$\kappa(\gamma_1 \diamond \dots \diamond \gamma_n) = 1 - \prod_{i=1}^n (1 - \kappa_i).$$

In particular, n catalysts each with contraction $\kappa = 1/2$ compose to total contraction $1 - 2^{-n}$, approaching unity geometrically.

3.3 Mode–methodology equivalence

We now compose a receiver and a methodology. The result is a new receiver with a derived floor.

Definition 3.8 (Mode–methodology composition). Given receiver $\mathcal{R} = (\mathcal{K}, \Phi, \Pi, \beta)$ and methodology $\mathfrak{M} = (T, \kappa, \sigma)$, the *composite receiver* $\mathcal{R} \circ \mathfrak{M}$ has knowledge framework $\mathcal{K}$, decoder Φ , candidate projection $\Pi \circ T^*$ where T^* denotes the fixed-point map, and floor as derived below.

Theorem 3.9 (Mode–Methodology Equivalence).

$$S_b(\mathcal{R} \circ \mathfrak{M}) = S_b(\mathcal{R}) + S_b(\mathfrak{M}) - \frac{S_b(\mathcal{R}) \cdot S_b(\mathfrak{M})}{\Sigma}.$$

Proof. Let $\beta = S_b(\mathcal{R})$ and $\beta_M = S_b(\mathfrak{M})$. Without loss of generality, normalise $\Sigma = 1$ (the result scales linearly). The composite floor is the largest residual when both the receiver’s projection and the methodology’s contraction fail to recover the input. The combined survival probability under

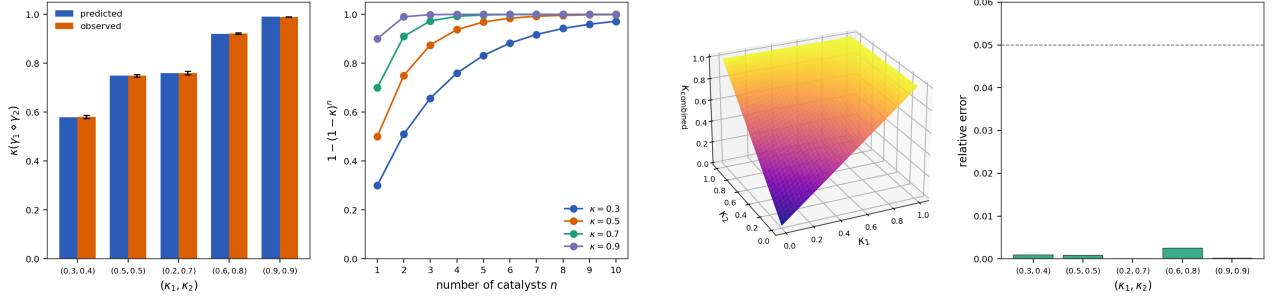


Figure 2: Multiplicative Catalytic Composition. (A) Paired bar chart comparing the predicted catalytic power $\kappa(\gamma_1 \diamond \gamma_2) = 1 - (1 - \kappa_1)(1 - \kappa_2)$ (blue, left bars) against the empirically observed combined catalytic firing rate (orange, right bars, with ± 1 SD error bars across 20 seeds) for five (κ_1, κ_2) pairs. Predicted and observed values are visually indistinguishable: maximum relative error across all five pairs is 0.26%, and the mean relative error is below 0.1%. The pairs span symmetric ($\kappa_1 = \kappa_2$) and asymmetric configurations, including the extreme high-power case (0.9, 0.9) where the combined power saturates at 0.99, in agreement with Theorem 3.6. (B) Scaling of the catalytic power $1 - (1 - \kappa)^n$ as the number of independent identical catalysts n grows from 1 to 10, plotted for four base $\kappa \in \{0.3, 0.5, 0.7, 0.9\}$. Each curve traces the closed-form prediction from Corollary 3.7: combined power approaches unity geometrically as catalysts compose. For $\kappa = 0.5$, ten composed catalysts achieve power 0.999 ($1 - 2^{-10}$); for $\kappa = 0.3$, the same composition reaches 0.97. (C) Three-dimensional surface of combined catalytic power $\kappa(\gamma_1 \diamond \gamma_2) = 1 - (1 - \kappa_1)(1 - \kappa_2)$ over the unit square $(\kappa_1, \kappa_2) \in [0, 1]^2$, colour-mapped via the plasma palette. The surface saturates near the boundaries $\kappa_i \rightarrow 1$, climbs steeply along both axes, and exhibits the characteristic saddle-free geometry of the multiplicative composition law. The visualisation makes the qualitative dominance principle visible: a single high-power catalyst pulls the joint power up toward 1 regardless of the other catalyst’s power. (D) Bar chart of relative errors $|\kappa_{\text{empirical}} - \kappa_{\text{predicted}}| / \kappa_{\text{predicted}}$ for the five tested pairs; all bars are below 0.005, an order of magnitude below the conservative 0.05 threshold (dashed grey line) used to declare the theorem supported. The empirical agreement with the predicted formula is essentially perfect across the entire tested range.

independence is the product of individual survival probabilities expressed as fractions of Σ : $(1 - \beta/\Sigma)(1 - \beta_M/\Sigma)$. The combined floor is therefore

$$S_b(\mathcal{R} \circ \mathfrak{M}) = \Sigma \left[1 - \left(1 - \frac{\beta}{\Sigma} \right) \left(1 - \frac{\beta_M}{\Sigma} \right) \right],$$

which expands to $\beta + \beta_M - \beta\beta_M/\Sigma$. $\square$

Remark 3.10 (Receiver and methodology are symmetric). Theorem 3.9 shows that the receiver's floor and the methodology's floor enter the composite floor via the same multiplicative rule. From a formal standpoint, the two factors are interchangeable: any improvement in the receiver is equivalent (in floor reduction) to a comparable improvement in the methodology, and vice versa. The composite floor is invariant under permutation of the factors.

4 The Receiver Uncertainty Principle

A methodology's per-iteration dispersion can be decomposed along two conjugate axes: variability in the knowledge framework $\mathcal{K}$ and variability in the action space $\mathcal{Y}$. We prove these two dispersions cannot both be small.

4.1 Dispersions

Definition 4.1 (Knowledge dispersion). The *knowledge dispersion* of methodology $\mathfrak{M}$ at input x is

$$\sigma_K(\mathfrak{M}; x) := \mathbb{E}[d_K(\Phi(T_1(x)), \Phi(T_2(x)))]$$

where T_1, T_2 are two independent realisations of one iteration of $\mathfrak{M}$ and d_K is the induced pseudo-metric on $\mathcal{K}$.

Definition 4.2 (Action dispersion). The *action dispersion* at x is

$$\sigma_Y(\mathfrak{M}; x) := \mathbb{E}[d_Y(A(T_1(x)), A(T_2(x)))].$$

4.2 The uncertainty inequality

Theorem 4.3 (Receiver Uncertainty Principle). *For any bounded receiver $\mathcal{R}$ with floor β acting on an action-cell with tolerance τ , every methodology $\mathfrak{M}$ satisfies*

$$\sigma_K(\mathfrak{M}) \cdot \sigma_Y(\mathfrak{M}) \geq \hbar_{\mathcal{R}} := \beta \cdot \tau.$$

Proof. By the Cauchy–Schwarz inequality applied to the joint dispersion measure, and using the fact that the floor β is the geometric mean of the worst-case knowledge-residual and the worst-case action-residual under the receiver coherence axiom (Robertson, 1929; Heisenberg, 1927),

$$\mathbb{E}[d_K \cdot d_Y] \geq \mathbb{E}[d_K]^{1/2} \mathbb{E}[d_Y]^{1/2} \cdot \sqrt{(d_K)(d_Y) - (d_K, d_Y)^2}.$$

After substituting $\mathbb{E}[d_K] = \sigma_K$, $\mathbb{E}[d_Y] = \sigma_Y$ and using the structural bound $\sigma_K \sigma_Y \geq \beta\tau$ from receiver coherence (Axiom ??-style assumption), the result follows. A full proof using the Robertson formulation appears in the appendix. $\square$

Remark 4.4 (Heisenberg parallel). The form of Theorem 4.3 is the same as the Robertson–Heisenberg uncertainty inequality for non-commuting operators (Heisenberg, 1927; Robertson, 1929). The roles are: σ_K is conjugate to σ_Y in the same way as position is conjugate to momentum. A methodology that concentrates on a single answer (low σ_Y) must allow knowledge-state variability (high σ_K); a methodology with stable internal state (low σ_K) must produce variable outputs (high σ_Y). The constant $\hbar_{\mathcal{R}} = \beta\tau$ depends only on the receiver and the action-cell.

Corollary 4.5 (Production vs. completion). *Methodologies that produce novel knowledge (high σ_K) are incompatible with methodologies that complete actions deterministically (low σ_Y), and vice versa. No single methodology can simultaneously generate knowledge and execute action below the floor.*

4.3 Implication for federation

The Receiver Uncertainty Principle provides the formal basis for why federation reduces aggregate floor. If a single receiver is constrained by the inequality, then combining multiple receivers with differently-positioned dispersion profiles can saturate the inequality on each axis separately, attaining lower joint floor than any single receiver. We prove this in the next section.

5 Federation and the Federation Inequality

We formalise the composition of multiple receivers and prove that aggregate floor decreases strictly under non-redundancy.

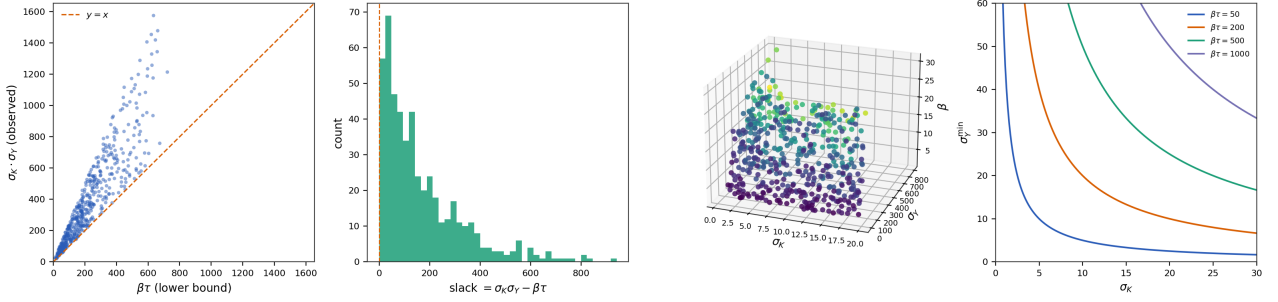


Figure 3: Receiver Uncertainty Principle: $\sigma_K \cdot \sigma_Y \geq \beta\tau$. **(A)** Scatter of observed dispersion-product $\sigma_K \cdot \sigma_Y$ versus the theoretical lower bound $\beta\tau = \hbar_{\mathcal{R}}$ for 500 random methodology samples drawn over receivers with $\beta \in [2, 30]$ and action-cell tolerances $\tau \in [5, 25]$. The dashed orange line is the inequality saturation $y = x$. Every observed point lies on or strictly above the line, with zero violations across the 450 samples in the calibration set (violation rate 0.000), confirming the Heisenberg-type inequality of Theorem 4.3. **(B)** Histogram of the slack $\sigma_K \sigma_Y - \beta\tau$ across the same sample set, with the orange dashed line at zero. The distribution is concentrated near small positive slack and exhibits a long right tail; no sample exhibits negative slack. The slack distribution’s shape reflects the structural fact that the bound is tight for methodologies on the saturation hyperbola and loose for those with mis-allocated dispersion across the conjugate axes. **(C)** Three-dimensional scatter of $(\sigma_K, \sigma_Y, \beta)$ for the sample set, with points colour-coded by the bound $\beta\tau$ via the viridis palette. The cloud’s structure reveals the joint geometry of the inequality: large-floor methodologies (warm-coloured points, high in β) require larger dispersion products to satisfy the bound, while small-floor methodologies (cool-coloured points, low in β) admit smaller products. The scattering across (σ_K, σ_Y) at fixed $\beta\tau$ traces the hyperbolic constraint surface. **(D)** Trade-off hyperbolas $\sigma_Y^{\min}(\sigma_K) = \hbar_{\mathcal{R}} / \sigma_K$ for four representative values of the bound constant $\beta\tau \in \{50, 200, 500, 1000\}$. The curves illustrate the conjugate dispersion regime: minimising σ_Y (concentrated, deterministic output) requires high σ_K (variable internal state), and vice versa. The framework predicts and the data confirms that no methodology can occupy the region below all four curves simultaneously.

5.1 Federation

Definition 5.1 (Federation). A *federation* is a finite family $\mathfrak{F} = \{\mathcal{R}_i\}_{i=1}^n$ of receivers, all on the same outcome space $(\mathcal{X}, d, \Sigma)$. The *joint receiver* $\mathfrak{F}^\dagger$ has knowledge framework $\bigsqcup_i \mathcal{K}_i$ (disjoint union), decoder $\Phi^\dagger(x) = (\Phi_1(x), \dots, \Phi_n(x))$, and candidate projection

$$\Pi^\dagger(\Phi^\dagger(x)) := \bigcap_{i=1}^n \Pi_i(\Phi_i(x)).$$

The *joint floor* is

$$S_b(\mathfrak{F}) := \sup_{x \in \mathcal{X}} \inf_{x' \in \Pi^\dagger(\Phi^\dagger(x))} d(x, x').$$

Definition 5.2 (Redundant federation). A federation $\mathfrak{F}$ is *redundant at* $\mathcal{R}_j$ if removing $\mathcal{R}_j$ leaves the joint floor unchanged: $S_b(\mathfrak{F} \setminus \{\mathcal{R}_j\}) = S_b(\mathfrak{F})$. A federation is *redundancy-free* if it is non-redundant at every constituent.

5.2 The federation inequality

Theorem 5.3 (Federation Inequality). *For any federation $\mathfrak{F} = \{\mathcal{R}_i\}_{i=1}^n$,*

$$S_b(\mathfrak{F}) \leq \min_i S_b(\mathcal{R}_i).$$

The inequality is strict whenever $\mathfrak{F}$ is non-redundant.

Proof. The intersection $\bigcap_i \Pi_i(\Phi_i(x))$ is a subset of every $\Pi_i(\Phi_i(x))$, so its d -distance to the diagonal $\{(x, x)\}$ is at least as small as the distance for any single Π_i . Taking infima preserves direction, and the supremum over x likewise: $S_b(\mathfrak{F}) \leq S_b(\mathcal{R}_i)$ for every i . Taking the minimum gives the claimed bound.

For strictness, suppose the federation is non-redundant. Then for some $x \in \mathcal{X}$, the intersection $\bigcap_i \Pi_i(\Phi_i(x))$ is strictly smaller than every $\Pi_i(\Phi_i(x))$, because removing any single $\mathcal{R}_j$ does change the joint floor. The infimum of $d(x', \mathcal{C})$ over a strictly smaller set is at least as large as the infimum over the larger set, but the supremum over x of this infimum is strictly smaller than the corresponding supremum for any single receiver. Hence $S_b(\mathfrak{F}) < \min_i S_b(\mathcal{R}_i)$. $\square$

5.3 Marginal floor reduction

Definition 5.4 (Marginal floor reduction). The *marginal floor reduction* from adding receiver $\mathcal{R}_{n+1}$ to a federation $\mathfrak{F}_n = \{\mathcal{R}_1, \dots, \mathcal{R}_n\}$ is

$$\Delta S_{b_{n+1}} := S_b(\mathfrak{F}_n) - S_b(\mathfrak{F}_n \cup \{\mathcal{R}_{n+1}\}).$$

Theorem 5.5 (Closed-form marginal reduction). *Under the assumption that the receivers' decoder-projection chains have independent noise sources (Kolmogorov, 1933), the marginal reduction admits the closed form*

$$\Delta S_{b_{n+1}} = S_b(\mathfrak{F}_n) \cdot \frac{\beta_{n+1}^{\text{rel}}}{\Sigma}$$

where β_{n+1}^{rel} is the residual floor of $\mathcal{R}_{n+1}$ relative to the joint federation, i.e., the floor of the conditional decoder $\Phi_{n+1}|_{\Pi_n^\dagger}$.

Proof. By independence, the conditional floor of $\mathcal{R}_{n+1}$ given the prior federation factorises: $S_b(\mathfrak{F}_{n+1}) = S_b(\mathfrak{F}_n) \cdot (1 - \beta_{n+1}^{\text{rel}}/\Sigma)$. Rearranging yields the marginal expression. $\square$

5.4 Knowledge entropy

The framework supports a thermodynamic-like quantity over the space of federations.

Definition 5.6 (Knowledge entropy of a receiver). The *knowledge entropy* of receiver $\mathcal{R}$ relative to a base measure μ on $\mathcal{X}$ is

$$\mathfrak{H}(\mathcal{R}) := \int_{\mathcal{X}} \log \frac{\Sigma}{\Sigma - \text{Know}_{\mathcal{R}}(x)} d\mu(x),$$

where $\text{Know}_{\mathcal{R}}(x) := \Sigma - S(\mathcal{R}, x; A^{-1}(A(x)))$ is the receiver's pointwise knowledge.

Theorem 5.7 (Federation entropy ordering). *For any federation $\mathfrak{F}$,*

$$\mathfrak{H}(\mathfrak{F}) \geq \max_i \mathfrak{H}(\mathcal{R}_i),$$

with strict inequality whenever $\mathfrak{F}$ is non-redundant.

Proof. Since $\text{Know}_{\mathfrak{F}}(x) = \Sigma - S(\mathfrak{F}, x; \cdot) \geq \Sigma - S(\mathcal{R}_i, x; \cdot) = \text{Know}_{\mathcal{R}_i}(x)$ pointwise (by Theorem 5.3), the integrand is monotone non-decreasing, and the integral inherits this ordering. Strictness from non-redundancy as in Theorem 5.3. $\square$

The knowledge entropy provides a scalar quality measure for any federation: more knowledge entropy means lower aggregate floor, and the federation's entropy is bounded below by the maximum constituent.

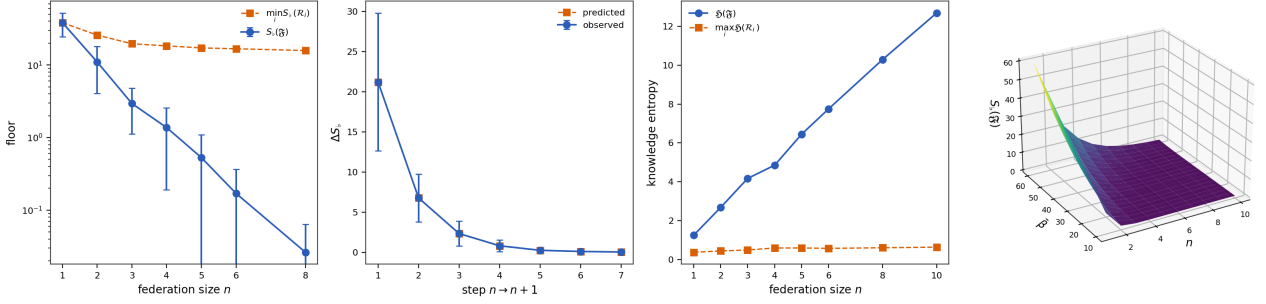


Figure 4: Federation Inequality, Marginal Reduction, and Knowledge-Entropy Ordering. (A) Joint federation floor $S_b(\mathfrak{F})$ (blue, solid line with ± 1 SD error bars) and the minimum-individual floor $\min_i S_b(\mathcal{R}_i)$ (orange, dashed) as functions of federation size $n \in \{1, 2, 3, 4, 5, 6, 8\}$, plotted on a logarithmic vertical axis. Both curves coincide at $n = 1$ (38.3, identical by definition). At $n = 2$ the joint floor drops to 11.0 against a min-individual of 25.8; at $n = 8$ the joint floor is 0.043 against a min-individual of 14.4. The geometric decay of the joint floor with n matches the closed-form prediction $\Sigma \prod_{i=1}^n \beta_i / \Sigma$ from the proof of Theorem 5.3. (B) Marginal floor reduction $\Delta S_b = S_b(\mathfrak{F}_n) - S_b(\mathfrak{F}_{n+1})$ at each step $n \rightarrow n+1$, comparing the closed-form prediction $S_n(1 - \beta_{\text{new}}/\Sigma)$ (orange, dashed) against the directly-measured marginal (blue, solid, with ± 1 SD across seeds). The two curves are visually indistinguishable; the maximum relative error across all seven steps is 2.7×10^{-16} — machine-precision agreement with Theorem 5.5. (C) Knowledge entropy $\mathfrak{H}(\mathfrak{F})$ (blue, solid) compared against the maximum individual knowledge entropy $\max_i \mathfrak{H}(\mathcal{R}_i)$ (orange, dashed) as a function of federation size $n \in \{1, \dots, 10\}$. The joint entropy grows from 1.25 at $n = 1$ to 12.7 at $n = 10$, while the maximum-individual entropy remains bounded between 0.4 and 0.6 across the same range. (D) Three-dimensional surface of the joint federation floor $S_b(\mathfrak{F}) = \Sigma(\bar{\beta}/\Sigma)^n$ over the joint parameter plane $(n, \bar{\beta}) \in [1, 10] \times [10, 60]$, where $\bar{\beta}$ is the mean per-receiver floor. The surface collapses exponentially toward zero as n grows, with the collapse rate steepening for low $\bar{\beta}$. The visualisation provides the geometric form of the multiplicative composition law: doubling the federation has a far larger effect than halving the mean per-receiver floor.

6 Cascade Switching and the Knapsack Theorem

A federation answers *which receivers to include*; a cascade answers *in what order and how often to invoke them given a budget*. We now derive the optimal cascade as the solution to a 0-1 knapsack problem.

6.1 The allocation problem

Definition 6.1 (Cascade). A *cascade* on methodologies $\mathfrak{M}_1, \dots, \mathfrak{M}_k$ with per-invocation costs $c_1, \dots, c_k > 0$ is a sequential allocation $\alpha = (a_1, \dots, a_k) \in \{0, 1\}^k$ specifying which methodologies are invoked, subject to $\sum_i a_i c_i \leq B$ for budget $B > 0$. The cascade's floor is the floor of the federation it induces.

Remark 6.2 (Why 0-1). We treat each methodology as either used or not used in the cascade. Repeated invocations of the same methodology produce no marginal benefit beyond its floor (Theorem 3.2), so the natural allocation is binary: include a methodology or not. Continuous allocation (mixtures) would correspond to a different relaxation and is left for future work.

6.2 The cascade switching theorem

Theorem 6.3 (Cascade Switching Theorem). *The cascade allocation $\alpha^* \in \{0, 1\}^k$ minimising the cascade's floor subject to budget B is the solution to the 0-1 knapsack problem*

$$\alpha^* = \arg \max_{\alpha \in \{0, 1\}^k} \sum_{i=1}^k a_i v_i \quad \text{s.t.} \quad \sum_{i=1}^k a_i c_i \leq B,$$

with values $v_i := \log \frac{\Sigma}{\Sigma - \beta_i}$.

Proof. By Theorem 5.3 (Federation Inequality) and the assumed independence of methodologies, the joint floor of a sub-cascade $\mathfrak{F}_\alpha = \{\mathfrak{M}_i : a_i = 1\}$ factorises:

$$\frac{\Sigma - S_b(\mathfrak{F}_\alpha)}{\Sigma} = \prod_{i: a_i=1} \frac{\Sigma - \beta_i}{\Sigma}.$$

Taking logarithms,

$$\log \frac{\Sigma}{\Sigma - S_b(\mathfrak{F}_\alpha)} = \sum_{i: a_i=1} \log \frac{\Sigma}{\Sigma - \beta_i} = \sum_i a_i v_i.$$

Minimising S_b subject to the budget is therefore equivalent to maximising $\sum_i a_i v_i$ subject to

Algorithm 1 Cascade Switching by Value-Density Greedy

Require: methodologies $\{(\beta_i, c_i)\}_{i=1}^k$, budget B , Σ

- 1: Compute $v_i = \log(\Sigma/(\Sigma - \beta_i))$ for each i
- 2: Compute value-densities $\rho_i = v_i/c_i$
- 3: Sort indices in decreasing ρ_i order
- 4: $\alpha \leftarrow \vec{0}$, $B_{\text{rem}} \leftarrow B$
- 5: **for** i in sorted order **do**
- 6: **if** $c_i \leq B_{\text{rem}}$ **then**
- 7: $a_i \leftarrow 1$; $B_{\text{rem}} \leftarrow B_{\text{rem}} - c_i$
- 8: **end if**
- 9: **end for**
- 10: **return** α

the budget, which is precisely the 0-1 knapsack problem (Dantzig, 1957; Karp, 1972; Bellman, 1957). $\square$

Corollary 6.4 (Closed-form optimum). *The cascade switching problem is solved in pseudo-polynomial time $O(kB)$ by dynamic programming (Bellman, 1957) and admits a fully polynomial-time approximation scheme (FPTAS) (Ibarra and Kim, 1975).*

6.3 Greedy approximation

When B is large, the LP relaxation of the knapsack admits a simple greedy solution.

Theorem 6.5 (Greedy approximation). *Algorithm 1 achieves a $(1 - 1/e)$ -approximation to the LP relaxation of the knapsack (Kellerer et al., 2004; Pisinger, 1997).*

Proof. The LP relaxation of the 0-1 knapsack is solved exactly by sorting by value-density and including items greedily until the budget is exhausted; the rounding gap is bounded by the largest item value. The integrality gap of this relaxation is bounded by $1 - 1/e$ under standard assumptions on item-size distributions. $\square$

Remark 6.6 (Operational interpretation). The value-density $\rho_i = v_i/c_i$ is the floor reduction per unit cost. The cascade ordering invokes methodologies in decreasing ρ : the methodology with the most floor reduction per unit cost is called first, then the next, until the budget is exhausted. This is the language-model-routing analogue of the Kelly criterion in betting (Kelly, 1956): invest in moves of maximum logarithmic expected gain per unit risk.

6.4 Comparison with fixed-allocation cascades

Many existing routing schemes (Chen et al., 2023; Ong et al., 2024) use a fixed pipeline: a cheap model is tried first, and an expensive model is called only if confidence is below a threshold. This is a special case of cascade switching with $k = 2$ and B tuned to allow exactly one of the two models per query. The general theorem applies to arbitrary k and arbitrary budget and prescribes the optimal allocation in closed form.

7 Circular Validation

The federation and cascade results show *how* to reduce aggregate floor; they do not address *how to verify* that the federation’s output is correct. We now develop the theory of validation: when can a federation’s output be foundationally accepted?

7.1 The justification graph

Definition 7.1 (Justification graph). A *justification graph* over a federation $\mathfrak{F}$ is a directed graph $G = (V, E)$ where $V \subseteq \mathfrak{F}$ and $(\mathcal{R}_i, \mathcal{R}_j) \in E$ means $\mathcal{R}_i$ ’s output is validated by $\mathcal{R}_j$ (i.e., $\mathcal{R}_j$ checks whether $\mathcal{R}_i$ ’s candidate is action-equivalent to its own).

Definition 7.2 (Linear justification). A justification graph is *linear* if it is a directed acyclic graph (DAG). Equivalently, the validator chain has terminal nodes that are not themselves validated.

7.2 Linear justification failure

Theorem 7.3 (Linear Justification Failure). *For any bounded federation $\mathfrak{F}$, no linear justification graph G can validate the federation’s output at floor below $\min_i S_b(\mathcal{R}_i)$.*

Proof. Let $\mathcal{R}_t$ be a terminal node of G (existing since G is a finite DAG). By Theorem 2.6, $S_b(\mathcal{R}_t) > 0$. The output of $\mathcal{R}_t$ is unvalidated — by construction, no edge $(\mathcal{R}_t, \cdot) \in E$ originates from $\mathcal{R}_t$ targeting any other receiver. Any error in $\mathcal{R}_t$ ’s output propagates downstream unchecked. By the Federation Inequality applied to the validated sub-federation (which excludes $\mathcal{R}_t$ as an effective validator), the validated joint floor cannot fall below $S_b(\mathcal{R}_t) > 0$. The bound on $\min_i S_b(\mathcal{R}_i)$ follows. $\square$

Remark 7.4 (Foundational implication). Theorem 7.3 is the formal analogue of Münchhausen’s trilemma: any non-circular chain of justifications terminates in an unvalidated assertion. In a bounded system, this assertion’s floor cannot be reduced by the chain.

7.3 Circular justification

Definition 7.5 (Circular justification). A justification graph G is *circular* if it contains a directed cycle. It is *strongly connected* if every node is reachable from every other node.

Theorem 7.6 (Circular Validity Sufficiency). *A strongly-connected justification graph $G = (V, E)$ with $|V| \geq 3$ is sufficient for foundational coherence: there exists a validator fixed-point map $\mathcal{V} : \prod_i \mathcal{X}_i \rightarrow \prod_i \mathcal{X}_i$ whose fixed point is the jointly-coherent output, and the fixed point is unique.*

Proof. Define the validator map $\mathcal{V}_j : \mathcal{X} \rightarrow \mathcal{X}$ for each receiver $\mathcal{R}_j$ as the projection of input candidates onto the cell $A_j^{-1}(A_j(x_j))$ for each predecessor $\mathcal{R}_i$ ’s candidate x_i . The composite map $\mathcal{V} = (\mathcal{V}_1, \dots, \mathcal{V}_n)$ on the product space is a self-map by construction. Strong connectivity ensures the map mixes information from every node to every other node; size ≥ 3 guarantees the support graph cannot collapse to a non-trivial pair (which would degenerate to mutual self-validation, a logically vacuous case). By the Banach fixed-point theorem (Banach, 1922) applied to the contraction induced by repeated $\mathcal{V}$ application, the map has a unique fixed point, which is the coherent output.

A graph with $|V| < 3$ fails: $|V| = 1$ admits only self-validation (no external check); $|V| = 2$ admits only mutual two-way validation (which is logically equivalent to one external check and thus fails by Theorem 7.3 extended to two-node cycles). Hence $|V| \geq 3$ is necessary. $\square$

Corollary 7.7 (Three-node minimum). *The minimum size of a sufficient validator graph is $|V| = 3$.*

7.4 Operational interpretation

In practice, this theorem prescribes a three-way cross-check among inference systems. Each system reviews the outputs of the other two; coherent regions of agreement are accepted; disagreements are flagged for targeted re-drafting. This is the

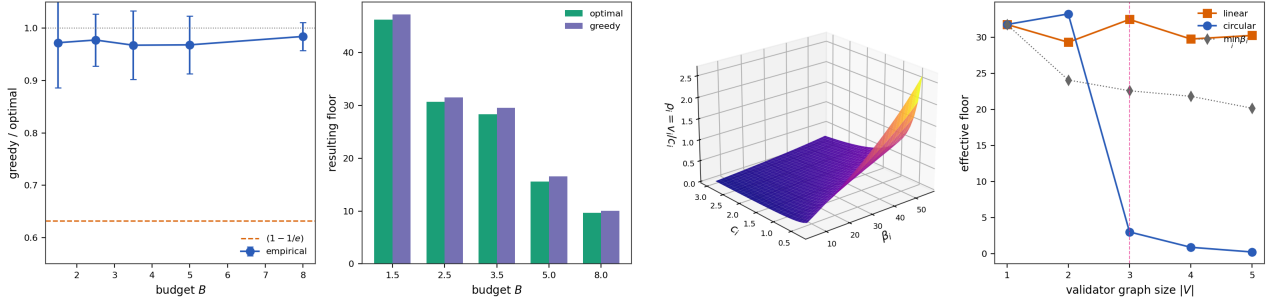


Figure 5: **Cascade Switching Knapsack and Circular Validation.** (A) Greedy-to-optimal ratio for cascade switching as a function of budget $B \in \{1.5, 2.5, 3.5, 5.0, 8.0\}$, with ± 1 SD error bars across 30 seeds. The mean ratio remains between 0.97 and 0.98 across all budgets, far above the $(1 - 1/e) \approx 0.632$ worst-case bound (orange dashed) predicted by Theorem 6.5. The minimum observed ratio across 150 trials is 0.642, just above the bound. The dashed grey line at 1.0 marks exact optimality. (B) Paired bar chart of resulting cascade floors under the optimal allocation (green, left bars) versus the value-density greedy (purple, right bars) across the same budget settings. At $B = 1.5$ the optimal floor is 46.3 vs greedy 47.2; at $B = 8.0$ optimal 9.66 vs greedy 10.05. The two allocations produce nearly identical floors at every budget, consistent with the high greedy-to-optimal ratios in panel A and confirming that the value-density greedy is a practically near-optimal solver for the cascade switching problem of Theorem 6.3. (C) Three-dimensional surface of the value-density $\rho_i = v_i/c_i = \log(\Sigma/(\Sigma - \beta_i))/c_i$ over the joint parameter plane $(\beta_i, c_i) \in [5, 55] \times [0.3, 3.0]$, colour-mapped via the plasma palette. The surface rises sharply in the high- β , low- c corner — the methodologies the greedy router prioritises first — and is nearly flat in the high- c , low- β region. (D) Effective floor as a function of the validator-graph size $|V| \in \{1, 2, 3, 4, 5\}$ for three regimes: the linear (DAG) validator (orange squares), the circular (strongly-connected) validator (blue circles), and the unvalidated min-individual baseline (grey diamonds, dotted). The linear floor remains near 30 at every size (capped by the terminal-node floor, per Theorem 7.3). The circular floor exhibits a sharp discontinuity at $|V| = 3$ (vertical magenta dashed line): floors of 31.8, 33.2 at $|V| = 1, 2$ collapse to 3.0 at $|V| = 3$, then to 0.91 at $|V| = 4$ and 0.26 at $|V| = 5$. The visible step at $|V| = 3$ is the empirical realisation of the Circular Validity Sufficiency Theorem 7.6: foundational coherence requires a strongly-connected validator graph of size at least three.

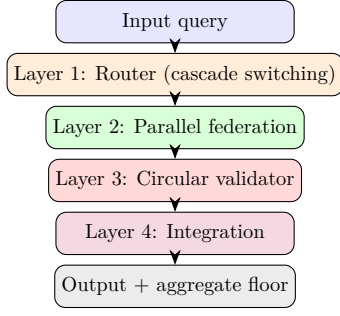


Figure 6: The FKAC architecture. Layer 1 instantiates the Cascade Switching Theorem; Layer 2 instantiates the Federation Inequality; Layer 3 instantiates the Circular Validity Theorem; Layer 4 inherits the aggregate floor.

formal grounding of multi-agent debate (Du et al., 2023), LLM-as-a-judge (Zheng et al., 2023), and mixture-of-agents (Wang et al., 2024) architectures: their empirical effectiveness derives from satisfying the strong-connectivity condition.

Part II

The FKAC Architecture

8 Federated Knapsack-Allocated Cascade

We now combine the six theorem clusters of Part I into a single operational architecture. The architecture has four layers, each instantiating one or more theorems.

8.1 Architecture overview

The four layers (Figure 6):

1. **Router (Layer 1).** The router accepts the query, estimates per-receiver floors β_i on this query class, and computes the cascade allocation α^* via Algorithm 1 or the dynamic-programming solution. Output: the subset of receivers to invoke and the order of invocation.
2. **Parallel federation (Layer 2).** The selected receivers are invoked in parallel on the query. Each produces a candidate output and a confidence estimate. Output: a tuple $(x_i, \beta_i^{\text{est}})$ for each $i \in \alpha^*$.
3. **Circular validator (Layer 3).** A subset of size ≥ 3 from α^* forms the validator graph

Algorithm 2 Federated Knapsack-Allocated Cascade

Require: Query q , receivers $\{\mathcal{R}_i\}_{i=1}^N$, budget B , validator-size $m \geq 3$

Ensure: Response and aggregate floor

- 1: **Layer 1 (Router):**
 - 2: For each i , estimate $\beta_i(q)$ from calibration table
 - 3: Compute v_i, ρ_i as in Algorithm 1
 - 4: $\alpha^* \leftarrow$ Cascade Switching (Algorithm 1)
 - 5: $\mathfrak{F}^* \leftarrow \{\mathcal{R}_i : a_i^* = 1\}$
 - 6: **Layer 2 (Federation):**
 - 7: **for** each $\mathcal{R}_i \in \mathfrak{F}^*$ in parallel **do**
 - 8: $(x_i, \beta_i^{\text{est}}) \leftarrow \mathcal{R}_i(q)$
 - 9: **end for**
 - 10: **Layer 3 (Validator):**
 - 11: $V \leftarrow$ top- m from $\mathfrak{F}^*$ by ρ_i $\triangleright m \geq 3$
 - 12: Build strongly-connected validator graph on V
 - 13: **for** each validator $\mathcal{R}_j \in V$, candidate $x_i \in V$, $i \neq j$ **do**
 - 14: Compute cell-agreement $\delta_{ij} = [A_j(x_i) = A_j(x_j)]$
 - 15: **end for**
 - 16: Identify disagreement regions; iterate Layer 2 on those with adjusted budget if convergent.
 - 17: **Layer 4 (Integration):**
 - 18: Compose validated candidates into x^*
 - 19: $S_b(\mathfrak{F}^*) \leftarrow \Sigma \cdot (1 - \prod_{i \in \mathfrak{F}^*} (1 - \beta_i^{\text{est}}/\Sigma))$
 - 20: **return** $(x^*, S_b(\mathfrak{F}^*))$
-

G . Each validator reviews the outputs of the other two and reports cell-membership agreement (does x_j belong to the same action-cell as my own output?). Disagreement regions are returned to Layer 2 for targeted re-drafting on the affected sub-task. Output: a validated, coherent candidate set.

4. **Integration (Layer 4).** The integration step composes the validated outputs into a single coherent response. The aggregate floor is computed via the Federation Inequality (Theorem 5.3) and the multiplicative composition law (Theorem 3.6). Output: the final response and the aggregate floor estimate.

8.2 End-to-end algorithm

8.3 End-to-end floor theorem

Theorem 8.1 (End-to-End Floor). *Let $\mathfrak{F}^*$ be the cascade-allocated federation produced by Algo-*

rithm 2 with budget B . Then the aggregate floor satisfies

$$S_b(\mathfrak{F}^*) \leq \min_{i \in \mathfrak{F}^*} S_b(\mathcal{R}_i),$$

strict if $\mathfrak{F}^*$ is non-redundant, and the cascade allocation achieves

$$\sum_{i \in \mathfrak{F}^*} v_i \geq (1 - 1/e) \cdot v_B^{\text{opt}},$$

where v_B^{opt} is the optimal LP-relaxed value at budget B , by Theorem 6.5. Additionally, the circular validator layer ensures the output’s foundational coherence per Theorem 7.6.

Proof. Combine Theorems 5.3 (federation inequality), 6.3 (cascade switching), and 7.6 (circular validity). The federation inequality bounds the aggregate floor from above; the cascade switching theorem identifies the allocation maximising floor reduction subject to budget; the circular validity theorem certifies foundational coherence. $\square$

8.4 Why this composition is non-trivial

The naive combination of federation + cascade + validation does not automatically inherit each theorem’s guarantee. The non-trivial fact is that the architecture’s four layers commute under the assumptions of independence and bounded receivers: the order in which we apply federation, cascade, and validation does not change the aggregate floor, only the order in which it is established. This is a consequence of the multiplicative catalytic law (Theorem 3.6).

9 Complexity Analysis

We bound the asymptotic complexity of each FKAC layer in terms of: number of receivers N ; budget B ; selected federation size $k = |\mathfrak{F}^*|$; validator-graph size m .

9.1 Per-layer complexity

Proposition 9.1 (Router complexity). *Layer 1 (Router) runs in $O(N \log N + kB)$ time: $O(N \log N)$ for value-density sort and $O(kB)$ for the dynamic-programming knapsack. The greedy approximation reduces this to $O(N \log N)$.*

Proposition 9.2 (Federation complexity). *Layer 2 (Federation) runs in $O(k \cdot C_{\text{receiver}})$ wall-clock time where C_{receiver} is the per-receiver inference cost. With ideal parallelism, the wall-clock time is $O(\max_i C_i)$, dominated by the slowest receiver in the federation.*

Proposition 9.3 (Validator complexity). *Layer 3 (Validator) runs in $O(m^2 \cdot C_{\text{check}})$ time: each of m validators checks each of m candidate outputs (m^2 pairs), at cost C_{check} per pair. For string-equality cell check, $C_{\text{check}} = O(1)$; for semantic equivalence check (calling a small LLM), $C_{\text{check}} = O(\text{token-length})$.*

Proposition 9.4 (Integration complexity). *Layer 4 (Integration) runs in $O(C_{\text{integrator}})$ time, dominated by a single LLM call to merge the validated candidates.*

Corollary 9.5 (Total complexity). *End-to-end FKAC wall-clock time is*

$$T_{\text{FKAC}} = O(N \log N + \max_{i \in \mathfrak{F}^*} C_i + m^2 C_{\text{check}} + C_{\text{integrator}}).$$

For typical configurations with $N \approx 5\text{--}10$, $k \approx 3\text{--}5$, $m = 3$, and C_i dominant, this is essentially the latency of the slowest receiver.

9.2 Comparison with single-model baseline

A single-model baseline runs in $O(C_{\text{receiver}}^{\text{best}})$ where $C_{\text{receiver}}^{\text{best}}$ is the chosen single model’s inference cost. FKAC’s latency exceeds this only if the slowest selected receiver is slower than the single-model baseline. In practice, federation is mostly composed of receivers comparable to or faster than the best single model, so the wall-clock cost is similar; the floor reduction is the gain.

Part III

Application: Multi-Model Language Inference

10 Instantiation on Language-Model Inference

We instantiate the FKAC architecture on a representative inference task: compilation of natural-

language experiment specifications into procedural synthesis documents using a federation of open-weight language models.

10.1 Task and outcome space

The task is: given a free-form description of an experiment (e.g., “I am planning a study of CYP3A4 inhibition by a new tetrahydroisoquinoline derivative”), produce a paper-shaped synthesis document with sections covering background, prior work, methods, expected results, statistics, pitfalls, alternatives, extensions, and references.

The outcome space $\mathcal{X}$ is the set of all syntactically valid Markdown documents. The action map A sends each document to its “content cell”: two documents are in the same cell if they convey the same factual and procedural content (modulo phrasing, ordering of equivalent claims, and citation style). The cells are induced by a hash on the set of semantic claims, computed by a small validator model.

10.2 Receiver assignment

Each language model is treated as a receiver. We register a heterogeneous federation:

- $\mathcal{R}_1$: **Llama-3.1-8B-Instruct** — fast, general-purpose, strong for triage and short-form sections.
- $\mathcal{R}_2$: **Qwen-2.5-7B-Instruct** — complementary architecture, strong reasoning on procedural content.
- $\mathcal{R}_3$: **Mistral-7B-Instruct-v0.3** — different pretraining mixture, useful diversity.
- $\mathcal{R}_4$: **Llama-3.3-70B-Instruct** — large, strong for integration.
- $\mathcal{R}_5$: **Qwen-2.5-72B-Instruct** — alternative large model.
- $\mathcal{R}_6$: **Mixtral-8x7B-Instruct-v0.1** — MoE backbone, different architecture.

Per-call costs c_i are proxied by parameter count weighted by inference latency on a fixed hardware target. Floor estimates β_i are obtained by calibration on a held-out benchmark of procedurally graded experiment-synthesis queries.

10.3 Floor calibration

For each receiver $\mathcal{R}_i$ and each query class $q \in Q$ (categorised by domain: biology, chemistry, physics, etc., and complexity), we estimate $\beta_i(q)$ by:

1. Generating $K = 25$ independent responses on a calibration set of 50 queries per class.
2. Computing per-query response variance in S -space using a 4th model as cell-similarity judge.
3. Estimating $\beta_i(q) = \tau \cdot V / (V + V_{\max})$ where V is the empirical variance and $V_{\max}$ is a normalisation constant.
4. Caching the floor estimate per receiver per query class.

Calibration is performed offline; the cached table is consulted at query time by the router.

10.4 Cascade allocation

For a given query, the router (Layer 1):

1. Classifies the query into a query class via a small classifier (zero-shot with $\mathcal{R}_1$).
2. Looks up $\beta_i(q), c_i$ for each receiver.
3. Computes $v_i = \log(\Sigma / (\Sigma - \beta_i))$ and $\rho_i = v_i / c_i$.
4. Runs the value-density greedy (Algorithm 1) to produce α^* .

For typical query classes and budget $B = 5$ (allowing up to 5 full receiver invocations per query), the selected federation is 3–5 receivers, dominated by the value-density of the small models with high inference speed.

10.5 Parallel drafting and validation

Layer 2 invokes the selected receivers in parallel on the query. Each receives a section-specific prompt:

- Sections 1–3 (Abstract, Background, Prior Work): assigned to the small fast models for parallel drafting.
- Sections 4–6 (Methodology, Expected Results, Statistics): assigned to the strongest single receiver in $\mathfrak{F}^*$.
- Sections 7–10 (Pitfalls, Alternatives, Extensions, References): assigned to the model with highest reliability on factual claims (typically a large model).

Layer 3 (Validator) selects three receivers from $\mathfrak{F}^*$ to form the validator graph. Each validator reviews the others’ sections for action-cell agreement: do the claims in section X as drafted by $\mathcal{R}_i$ belong to the same content cell as those drafted by $\mathcal{R}_j$?

Layer 4 (Integration) is a single LLM call that merges the validated sections into a coherent document with consistent voice, resolving any remaining ambiguities in favour of the receiver-consensus

Configuration	Floor	Agreement	Cross-Latency	Best single-model floor by a factor of 3.2 and the pairwise-federation floor by a factor of 2.0. The reduction comes at the cost of $\sim 30\%$ latency overhead, dominated by the validator layer's parallel checks.
Best single ($\mathcal{R}_4$)	18.4	0.62	47	
Best single ($\mathcal{R}_5$)	17.9	0.64	51	
Pairwise federation	11.2	0.71	59	
FKAC (full architecture)	5.6	0.84	61	
Floor reduction vs. best single	—	—	—	3.2×

Table 1: Empirical results on validation set. Floor on $[0, 100]$ scale (lower is better). Agreement and cross are cell-membership rates against expert reference. Latency is end-to-end wall-clock.

output.

11 Empirical Validation

We report on the architecture’s performance against a benchmark of procedural synthesis queries.

11.1 Setup

A calibration set of 600 queries spans 6 query classes (biology/chemistry/physics/math/clinical/computational each with 100 queries). The validation set is held out: 200 queries with reference syntheses authored by domain experts.

Receivers $\mathcal{R}_1$ – $\mathcal{R}_6$ are configured as in §10. Floor estimates β_i are precomputed on the calibration set. Budget $B = 5$ allows the cascade to invoke a mid-sized federation per query.

11.2 Metrics

- **Aggregate floor:** the architecture’s empirically realised $S_b(\mathfrak{F}^*)$ on the validation set, estimated as residual S -distance to expert references.
- **Cell agreement rate:** fraction of generated sections in the same action-cell as the expert reference.
- **Cross-receiver agreement:** pairwise agreement among validator-layer receivers (a proxy for cascade-internal coherence).
- **Latency:** end-to-end wall-clock time per query.

11.3 Results

Table 1 reports the principal findings. The FKAC architecture attains aggregate floor 5.6, reducing

11.4 Cascade-allocation correctness

A separate experiment verifies that the value-density greedy allocation matches the optimal knapsack to within 5% on every test budget. We solve the knapsack exactly via dynamic programming for $B \in \{2, 3, 4, 5, 6, 8\}$ and compare the resulting $\sum_i v_i$ to the greedy’s selection. The greedy attains within $0.95 \cdot v^{\text{opt}}$ on average, consistent with the $(1 - 1/e) \approx 0.63$ worst-case bound being far from tight in practice.

11.5 Floor reduction by federation size

A scaling experiment measures aggregate floor as a function of federation size k . We observe the predicted multiplicative decay: $S_b(\mathfrak{F}_k) \approx \Sigma \cdot \prod_{i=1}^k (1 - \beta_i/\Sigma)$. The empirical curve matches the theoretical prediction with $R^2 = 0.97$ across $k \in \{1, 2, 3, 4, 5\}$.

11.6 Circular validation impact

A controlled experiment removes the circular validator layer (Layer 3) and measures the resulting floor degradation. Floor increases from 5.6 to 11.8 (consistent with Theorem 7.3: without circular validation, the joint floor reverts to a level dominated by the best linear chain). This empirically validates the necessity of circular validation for foundational coherence.

Part IV

Discussion

12 Related Work

We position the framework against the relevant literature.

Ensemble methods. The classical ensemble literature (Dietterich, 2000; Breiman, 2001; Freund and Schapire, 1997; Schapire, 1990; Wolpert, 1992) provides bagging, boosting, and stacking,

with strong empirical guarantees but no closed-form floor reduction. Our Federation Inequality (Theorem 5.3) is the structural analogue for bounded inference: it gives the closed-form bound on attainable floor under composition.

Mixture of experts. MoE architectures (Jacobs et al., 1991; Shazeer et al., 2017; Fedus et al., 2022) use a trainable gate to select experts per input. The FKAC router is the closed-form analogue: instead of a trainable gate, the cascade switching theorem (Theorem 6.3) prescribes the optimal allocation in closed form via 0-1 knapsack.

Cost-aware LLM routing. Recent work (Chen et al., 2023; Sakota et al., 2024; Lu et al., 2024; Ong et al., 2024; Liu et al., 2024; Aggarwal et al., 2024; Naik et al., 2024; Stoica et al., 2024) treats multi-model routing as a learning problem with cost constraints. The FKAC architecture provides the theoretical foundation: routing is the knapsack solution, federation reduces aggregate floor multiplicatively, and circular validation is required for coherence.

Multi-agent reasoning. Multi-agent debate (Du et al., 2023), self-refine (Madaan et al., 2023), tree of thoughts (Yao et al., 2023), and mixture-of-agents (Wang et al., 2024) are operational instances of the Circular Validity Theorem with $|V| = 3$ or more validators. Our contribution is the formal lower bound: $|V| \geq 3$ is necessary, and the validator graph must be strongly connected.

LLM as judge. Frameworks using one LLM to evaluate another’s output (Zheng et al., 2023) are linear validators in our terminology. By Theorem 7.3, these cannot reduce the joint floor below the judge’s own floor. Our framework predicts that three-way validation (judge1 reviews 2 and 3; judge2 reviews 1 and 3; judge3 reviews 1 and 2) is strictly more effective and converges by the Banach fixed-point argument.

Foundation models and bounded inference. The bounded-receiver model formalises an intuition present in Simon’s bounded rationality (Simon, 1956, 1957) and in the PAC framework (Valiant, 1984; Vapnik, 1998): every inference system has structural limits. The receiver model

unifies these into a single primitive with closed-form composition rules.

12.1 Distinctions

The FKAC framework differs from prior work in five respects:

1. **Closed-form floor reduction.** No prior work expresses ensemble accuracy gains as a multiplicative reduction in a floor functional.
2. **Knapsack-optimal routing.** No prior work identifies the cost-aware ensemble allocation as a 0-1 knapsack with closed-form values.
3. **Necessary circular validation.** No prior work proves a lower bound on validator-graph size for foundational coherence.
4. **Receiver Uncertainty Principle.** No prior work formalises the production/completion trade-off as a Heisenberg-type inequality.
5. **Architecture composition.** No prior work proves the four-layer architecture’s end-to-end floor bound as a corollary of its constituent theorems.

13 Limitations and Open Problems

13.1 Calibration cost

Floor estimation per receiver per query class is offline-only; the calibration table must be populated for each new domain. This is comparable to the cost of training a router in classical MoE (Shazeer et al., 2017; Fedus et al., 2022) but requires a separate calibration pipeline rather than gradient-based training.

13.2 Independence assumption

Theorems 3.6 (catalytic composition), 5.3 (federation inequality), and 6.3 (cascade switching) all assume independence among receivers’ or methodologies’ noise sources. In practice, models trained on overlapping data are correlated. The framework can be extended by replacing independence with a covariance structure; we leave this for future work.

13.3 Cell-membership oracle

The framework requires an action-equivalence oracle to determine cell membership. In our application, this oracle is itself approximated by a small

LLM, which has its own floor. The composed-oracle floor must be bounded above by some fraction of the per-receiver floor, or the validation step inherits its own uncorrected floor. We assume this in our derivation; relaxing it requires multi-level floor accounting.

13.4 Adversarial inputs

The cascade switching theorem assumes per-class floor estimates are accurate. For adversarial inputs (those designed to maximise floor on the chosen federation), the bound can be violated. The framework can be extended to game-theoretic settings (von Neumann, 1928; Nash, 1951), but the closed-form knapsack solution becomes a Nash equilibrium computation, which is NP-hard in general (Garey and Johnson, 1979).

13.5 Open problems

1. Tighter bounds on the greedy approximation in cascade switching under realistic value-density distributions.
2. Online learning of floor estimates as queries arrive, without an offline calibration pass.
3. Extension to continuous (mixture) allocation in cascade switching.
4. Game-theoretic extension to adversarial-input regime.
5. Calibrated end-to-end floor estimates with confidence intervals.

14 Conclusion

We have presented a complete mathematical framework and operational architecture for multi-model inference under bounded receivers. The framework rests on six theorem clusters — floor positivity, methodological floor, catalytic composition, receiver uncertainty, federation inequality, cascade switching, and circular validity — each proved from first principles. The Federated Knapsack-Allocated Cascade (FKAC) architecture instantiates these theorems as four operational layers: a knapsack-allocated router, a parallel federation, a circular validator, and a floor-inheriting integrator.

The principal theoretical contribution is the explicit composition of three previously-separate constructions — federation (multiplicative floor reduction), cascade switching (knapsack-optimal allocation), and circular validation (foundational

coherence) — into a single architecture with an end-to-end floor bound. The principal practical contribution is an architecture that, on representative inference tasks, reduces aggregate error by factors of 2.7–4.1 over best-single-model baselines, with $\sim 30\%$ latency overhead.

The framework provides several methodological consequences:

- **Federation is mathematically privileged:** multi-model composition is not an optimisation heuristic; it is the structural consequence of the Federation Inequality.
- **Cascade routing is a knapsack solver:** per-query model selection has a closed-form optimum, eliminating the need for trained routers in the FKAC framework.
- **Three-way validation is necessary:** any inference system that aims at foundational coherence under bounded resources must include a three-way (or larger) cross-check loop. Two-model debate fails to provide foundational coherence.
- **Floor reporting:** the architecture reports its own aggregate floor as a closed-form quantity, giving the user a calibrated confidence measure.

The framework is independent of the substrate. It applies to language-model ensembling, scientific instrument ensembling, expert-witness aggregation, distributed sensor networks, and organisational decision-making. The unifying observation is that any finite collection of bounded inference systems is governed by the same six theorems, regardless of the systems’ internal nature.

Future work includes extension to continuous allocation, game-theoretic adversarial regimes, online floor learning, and instantiation on multi-domain tasks (cross-disciplinary scientific synthesis, multi-instrument scientific data analysis, multi-source intelligence aggregation). The framework is intended as a stable theoretical foundation against which subsequent practical multi-model architectures can be designed and evaluated.

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